# SRI KRISHNA INSTITUTE OF TECHNOLOGY

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## Department of Artificial Intelligence and Machine Learning

**Subject Name: Analysis and Design of Algorithms** **Subject Code: BCS401**

**SEM: 4th** **DIV: A**

**Faculty: Prof. Manzoor Ahmed**

### Module-1 Question Bank

| SL# | Question | CO | Level | Marks |
|---|---|---|---|---|
| 1. | Define algorithm. Explain asymptotic notations Big Oh, Big Omega and Big Theta notations | CO1 | L2 | 08 |
| 2. | Explain the general plan for analyzing the efficiency of a recursive algorithm. Suggest a recursive algorithm to find factorial of number. Derive its efficiency | CO1 | L2 | 08 |
| 3. | Explain general plan of mathematical analysis of non-recursive algorithms. Design the algorithm to find Element Uniqueness Problem and obtain the time efficiency. | CO1 | L2 | 08 |
| 4. | If $t1(n) \in O(g1(n))$ and $t2(n) \in O(g2(n))$, then show that $t1(n) + t2(n) \in O(\max\{g1(n), g2(n)\})$. | CO1 | L2 | 04 |
| 5. | With neat diagram explain different steps in designing and analyzing an algorithm. | CO1 | L2 | 08 |
| 6. | Explain the general plan for analyzing the efficiency of a non-recursive algorithm. Suggest a non-recursive algorithm to find maximum element in the list of n numbers. Derive its efficiency | CO1 | L2 | 08 |
| 7. | With the algorithm derive the worst case efficiency for Bubble sort. | CO1 | L2 | 04 |
| 8. | What is an algorithm? Explain the fundamentals of algorithmic problem solving. | CO1 | L2 | 10 |
| 9. | Develop an algorithm to search an element in an array using sequential search. Calculate the best case, worst case and average case efficiency of this algorithm. | CO1 | L2 | 10 |
| 10. | Explain asymptotic notations with example. | CO1 | L2 | 10 |
| 11. | Give the general plan for analyzing the efficiency of the recursive algorithm. Develop recursive algorithm for computing factorial of a positive number. Calculate the efficiency in terms of order of growth. | CO1 | L2 | 10 |
| 12. | For each of the following functions, indicate how much the functions value will change if its argument is increased Fourfold.<br>a. $\log 2 n$ b. $\sqrt{n}$ c. $n$ d. $n^2$ e. $n^3$ f. $2 n$ | CO1 | L1 | 05 |
| 13. | For each of the following pairs of functions, indicate whether the first function of each of the following pairs has a lower, same, or higher order of growth (to within a constant multiple) than the second function. | CO1 | L1 | 05 |

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|---|---|---|---|---|
| | a. $n(n + 1)$ and $2000n^2$ b. $100n^2$ and $0.01n^3$<br>c. $\log_2 n$ and $\ln n$ d. $\log_2^2 n$ and $\log_2 n^2$<br>e. $2^{n-1}$ and $2^n$ f. $(n-1)!$ and $n!$ | | | |
| 14. | Define Order of growth? Explain basic efficiency classes and also arrange the same in increasing order of their magnitudes. | **CO1** | **L2** | **05** |
| 15. | Express the following using Big Oh, Big Omega and Big theta Notations.<br>i) $10n^3 + 5$ ii) $6*2^n + n^2$ | **CO1** | **L2** | **05** |
| 16. | Illustrate mathematical Analysis of recursive algorithm for Towers of Hanoi. | **CO1** | **L2** | **08** |
| 17. | Illustrate mathematical analysis of recursive algorithm for finding a number of binary digits in a decimal integer. | **CO1** | **L2** | **08** |
| 18. | Illustrate mathematical analysis of recursive algorithm for computing Fibonacci numbers. | **CO1** | **L2** | **08** |
| 19. | What is Pattern Matching? Write the algorithm for Pattern Matching and give the detailed analysis. Given a text string: FUN-UNCLE, pattern string: UNCLE, apply the above algorithm to the given data and show the steps. | **CO1** | **L2** | **08** |
| 20. | What is Brute Force method? Write the algorithm for Selection Sort and give its analysis. Apply the same for the given set of data and show the steps of sorting 45,20,40,45,15. | **CO1** | **L2** | **08** |
| 21. | Write the algorithm for Bubble Sort and give its analysis. Apply the same for the given set of data and show the steps of sorting 40,50,30,20,10. | **CO1** | **L2** | **08** |
| 22. | Write the algorithm for Sequential Search or Linear Search. | **CO1** | **L2** | **08** |
| 23. | What is Brute Force method? Explain the Properties of Sorting and Advantages and Disadvantages of Brute Force method. | **CO1** | **L2** | **08** |

**Faculty Signature**